

Machine Learning Approaches for Regional Carbon Flux Prediction: A Systematic Review

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Abstract

Accurate prediction of regional carbon fluxes is critical for climate mitigation policy, yet current process-based models carry large uncertainties. We review 84 studies applying machine learning to carbon flux estimation from 2015-2024, covering eddy covariance, satellite remote sensing, and atmospheric inversion datasets. Deep learning methods outperform traditional models by 23% on held-out test sets, but exhibit poor generalization across biomes and climate regimes not seen during training. We find that soil carbon dynamics, permafrost feedback, and extreme weather events are systematically underrepresented in training data, creating dangerous blind spots for climate projections beyond 2050. We propose a hybrid physical-ML framework as a path toward generalizable carbon cycle modeling.

1. Introduction

The global carbon cycle determines the pace of climate change. Terrestrial ecosystems absorb approximately 30% of annual anthropogenic CO₂ emissions, but this land carbon sink shows high interannual variability driven by temperature, precipitation, and disturbance regimes. Process-based land surface models (LSMs) disagree by up to 2 PgC/yr in their estimates of the tropical carbon balance, limiting confidence in climate projections.

Machine learning offers an alternative data-driven approach that can leverage the explosion of Earth observation data: eddy covariance flux towers (FLUXNET, 200+ sites), MODIS/VIIRS satellite products, and atmospheric CO₂ from OCO-2/3. However, the data-driven paradigm raises fundamental questions about out-of-distribution generalization and physical consistency that have not been systematically addressed.

2. Review Methodology

We searched Web of Science and Scopus for studies combining machine learning with carbon flux estimation, published January 2015 - June 2024. Inclusion criteria: (1) quantitative comparison with a baseline model, (2) evaluation on held-out temporal or spatial data, (3) focus on net ecosystem exchange (NEE), gross primary production (GPP), or soil respiration (Rs). After screening, 84 studies met criteria.

Studies were categorized by: ML method (random forest, gradient boosting, LSTM, CNN, transformer), input data source (in-situ, satellite, reanalysis), spatial scale (site, regional, global), and validation approach (temporal split, spatial cross-validation, leave-one-site-out).

3. Performance Analysis

Across all 84 studies, deep learning methods (LSTM, transformer, CNN) outperformed random forest and gradient boosting by 23% mean RMSE reduction on held-out test sets. Transformers showed the strongest performance for multi-site generalization ($R^2 = 0.87$ vs 0.79 for LSTM) likely due to their ability to model long-range temporal dependencies.

However, performance degrades sharply under spatial cross-validation: models trained on temperate forests predict tropical forest fluxes with $R^2 = 0.31$, compared to $R^2 = 0.84$ in-distribution. Arctic and boreal sites show the worst out-of-distribution performance ($R^2 = 0.19$), despite being critical for understanding permafrost carbon feedbacks.

Only 12 of 84 studies reported performance under climate extremes (drought, heatwave). In these studies, ML model error increased 2.8x during extreme events compared to baseline conditions, suggesting models have not learned the mechanisms driving flux anomalies under novel climate forcing.

4. Data Gaps and Blind Spots

Three systematic gaps in training data limit current ML carbon models. First, soil carbon dynamics: only 8% of studies included deep soil carbon (>30 cm) measurements, yet permafrost soils store ~1,500 PgC · roughly double the current atmospheric carbon stock. Second, extreme events: FLUXNET tower coverage of drought years is sparse and biased toward European sites, limiting generalization to tropical and monsoon systems.

Third, disturbance regimes: fire, insect outbreaks, and logging create large transient carbon fluxes poorly captured by satellite indices. Post-disturbance recovery trajectories lasting decades are absent from most training datasets, leading models to systematically underestimate forest regrowth carbon uptake by 15-40% in remote sensing studies.

We identify a critical need for targeted data collection in data-sparse high-impact regions: the Congo Basin, Siberian larch forests, and South American cerrado.

5. Toward Hybrid Physical-ML Models

Pure data-driven models lack the physical constraints needed for reliable extrapolation to future climate states. We propose a hybrid framework: use ML to learn residuals of a process-based LSM, constrained to conserve mass and energy. This approach, tested in 3 studies reviewed, achieves 18% lower RMSE than either the LSM or ML alone while maintaining physical plausibility.

Key open problems for hybrid models include: (1) differentiable LSM implementations that allow gradient-based training, (2) uncertainty quantification that propagates both model and observation uncertainty, (3) multi-objective training that balances predictive accuracy with physical constraint satisfaction. Neural ODEs and physics-informed neural networks represent promising architectures not yet applied to carbon cycle modeling.

6. Conclusion

Machine learning has substantially improved carbon flux prediction accuracy, but generalization across biomes and climate regimes remains poor. The systematic underrepresentation of permafrost, disturbance, and extreme events in training data poses risks for climate projections. Hybrid physical-ML architectures, combined with targeted data collection in underrepresented ecosystems, offer the most promising path toward reliable global carbon cycle modeling.

References

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